mgf

$E[e^{tX}]$

Order Statistics — important.

$Y_{(1)} = \min(Y_1, \dots Y_n)$

$Y_{(n)} = \max(Y_1 \dots Y_n)$

Chapter 7

Sampling dist. related to $N(\mu, \sigma^2)$

Thm 1: $Y_1 \dots Y_n$ iid sample $\sim N(\mu, \sigma^2)$

$$\left(\frac{\sum_{i=1}^{n}(Y_i - \bar{Y})}{\sigma}\right)^2 \sim \chi^2_n$$

Thm 2: $Y_1 \dots Y_n \sim$ iid $\sim N(\mu, \sigma^2)$

$$S^2 = \frac{1}{(n-1)} \sum_{i=1}^{n} (Y_i - \bar{Y})^2$$

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$$

s^2 and $\bar{Y}$ are independent

CLT

$$Y_1, \ldots Y_n \overset{iid}{\sim} Y$$

$$\frac{\sqrt{n}(\bar{Y}-\mu)}{\sigma} \xrightarrow[n\to\infty]{} N(0,1)$$

Chapter 8

Unbiased estimator

$N(\mu, 4)$

$\bar{Y}$ is estimator of μ

Unbiased - $E[\hat{\theta}] = \theta$

$$MSE(\hat{\theta}) = B(\hat{\theta})^2 + V[\hat{\theta}]$$

Sample variance $Y_1, \ldots Y_n \sim Y$

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (Y_i - \bar{Y})^2$$

S^2 is unbiased estimator of σ_Y^2

<u>Consistency</u>

$$\lim_{n \to \infty} P(|\hat{\theta} - \theta| > \varepsilon) = 0$$

or

$$\lim_{n \to \infty} P(|\hat{\theta} - \theta| < \varepsilon) = 1$$

if $\hat{\theta}$ is unbiased $\lim_{n \to \infty} V[\hat{\theta}_n] = 0$

$\therefore$ $\hat{\theta}$ is consistent.